

Scalability & Future Plan

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Team ID	SWTID-2026-7991
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan

This document outlines how EduGenie can be scaled to support larger numbers of learners, increased AI requests, and additional learning features in the future.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Dependence on external Gemini API availability	AI responses may be affected by network, service or quota limitations.	High
2	Response time can vary with request size and service load	Learners may experience slower responses during heavy usage.	Medium
3	Limited learning analytics in the current version	Less visibility into learner usage and progress.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Designed for normal project/demo usage	Use scalable hosting, request queuing and load balancing for larger user bases.
2	Data Storage	Minimal / optional persistent storage	Introduce a scalable database for profiles, learning history and analytics.
3	Performance	AI response depends on API/network	Use caching where appropriate, asynchronous processing and performance monitoring.
4	Security	Environment-based API configuration	Add authentication, authorization, secure secret management and stronger validation.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve UI, validation and testing coverage	Next release	Better usability and reliability
Phase 2	Add learner accounts, history and analytics	Future release	Personalized learning experience
Phase 3	Add multilingual and subject-specific learning modes	Future release	Broader learner support
Phase 4	Scale deployment and add advanced AI learning features	Long-term	Support larger usage and richer learning workflows